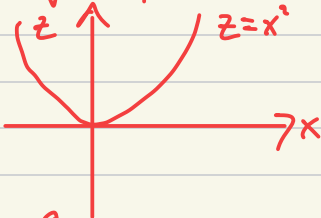


e.g. Graph $f(x, y) = x^2$

Let's ignore y



this $\uparrow$ lives on the plane $y = c$

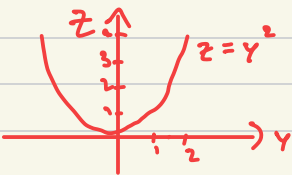


Graph sections: is a "slice" of the graph of $f(x, y) = z$ that is cut along a plane of constant x (or constant y)

idea: if we know how each cross-section looks like we can stack them together to graph the function

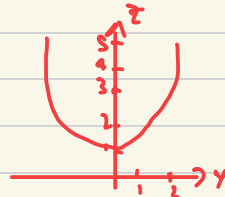
e.g. Graph $f(x, y) = x^2 + y^2$ by taking sections at $x = c$.

section: $x = 0 \leadsto f(0, y) = 0^2 + y^2 = z$

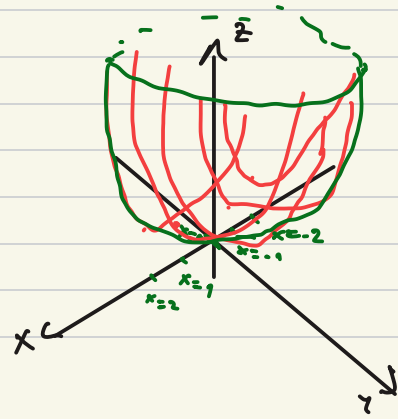


section $x = 1$

$$f(1, y) = 1 + y^2 = z$$



for arbitrary $x = c$
all we do is shift up $z = y^2$ by x^2



e.g. Graph $f(x,y) = xy$

level curves

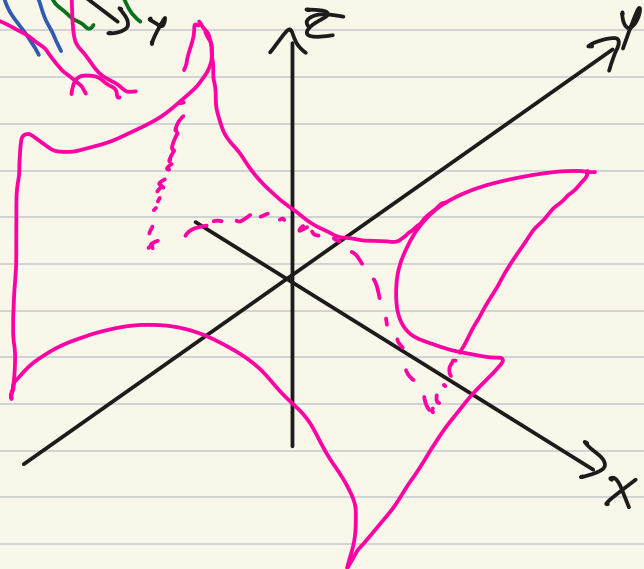
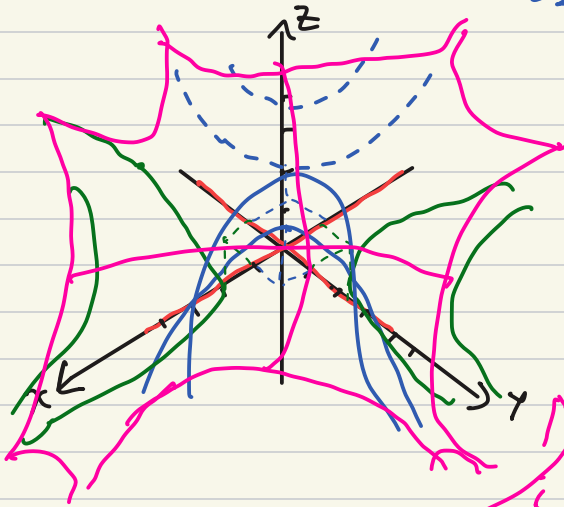
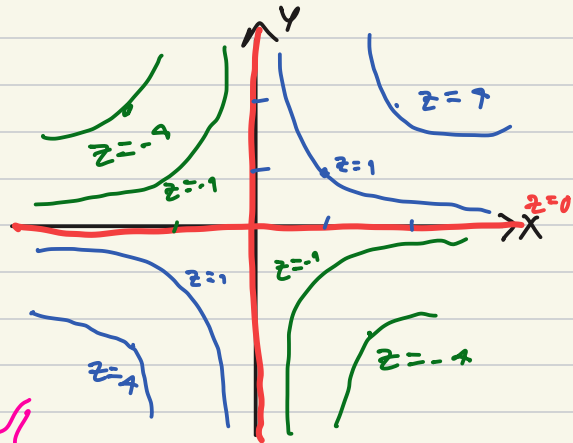
$$z=0 \rightarrow \{x=0\} \cup \{y=0\}$$

$$z=1 \rightarrow xy=1 \quad y=\frac{1}{x}$$

$$z=-1 \rightarrow xy=-1 \quad y=-\frac{1}{x}$$

$$z=4 \rightarrow xy=4 \quad y=\frac{4}{x}$$

$$z=-4 \rightarrow y=-\frac{4}{x}$$



e.g. Can we graph $f(x, y, z) = x^2 + y^2$

3 independent variables

1 component

explicit graph

lines inside $\mathbb{R}^3 \times \mathbb{R} = \mathbb{R}^4$

we can graph level sets

$$f = 0 \leadsto \{(0, 0, z) : z \in \mathbb{R}\}$$

$$f = 1 \leadsto x^2 + y^2 = 1 \quad z = \text{anything}$$

$$f = 4 \leadsto x^2 + y^2 = 4 \quad z \in \mathbb{R}$$

$$f = 9 \leadsto x^2 + y^2 = 9 \quad z \in \mathbb{R}$$

